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MACADITA.

Item one

$A = 40m^2$

- Speed above wings, $V_1 = 100 m/s$.
- Speed below wings, $V_2 = 120 m/s$.
- $\rho_{air} = 1.29 kg m^{-3}$.
- Aircraft mass, $m = 1500kg$.
- $g = 9.81 ms^{-2}$.

a) Using Bernoulli's Principle

$$P_1 + \frac{1}{2}\rho v_1^2 = P_2 + \frac{1}{2}\rho v_2^2.$$

$$P_1 - P_2 = \frac{1}{2}\rho(v_2^2 - v_1^2)$$

$$= \frac{1}{2} \times 1.29 (120^2 - 100^2)$$

$$= 0.645(14400 - 10000)$$

$$= 2838 Pa.$$

$$F = PA.$$

$$= 2838 \times 40$$

$$= 1.1352 \times 10^5 N$$

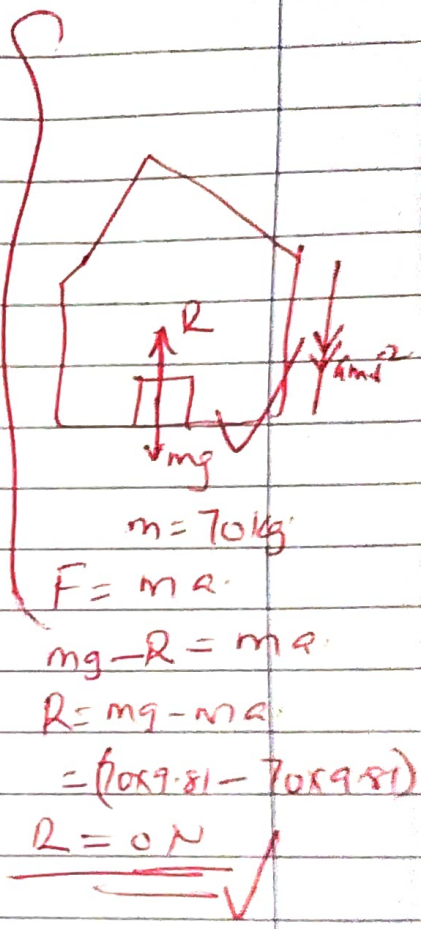
Weight of aircraft.

$$W = mg.$$

$$= 15000 \times 9.81$$

$$= 147150$$

$$= 1.4715 \times 10^5 N$$



m = 70kg

F = ma.

mg − R = ma

R = mg − ma

= (70 × 9.81 − 70 × 9.81)

R = 0 N

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flex - "soft landing"
not flexing - "hard landing"

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Since W > F, lift force is not sufficient for take off.

b) → Explanation of take off using Bernoulli's principle.
- An aircraft has a propelor which makes air molecules above wings with higher velocity, creating low pressure zone. Air molecules below wings have lower velocity, hence a higher pressure zone. Difference in pressure creates a net force called lift force necessary for take off.

c) → Musa felt weightless after the parachute developed a puncture because he entered free fall.
- During free fall $a = g$, the only force acting on him was his weight. His apparent weight, $R = 0$.
- Therefore, he experienced weightlessness.

d) → Musa flexed his knees, for him to have a soft landing ($Ft = I$). This increases the time of impact on the ground (impulse) and therefore reduces injuries.
- other passengers who never flexed had hard landing. Therefore time of impact was small and more pain was felt.

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$m = 2000kg$

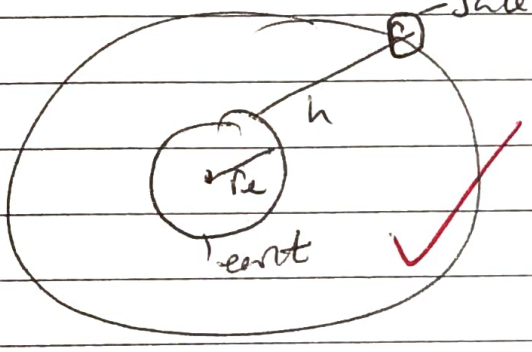
$R_e = 6.4 \times 10^6 m$

$h = 800km = 8.0 \times 10^5 m$

$R = r_e + h$

$= 6.4 \times 10^6 + 8.0 \times 10^5$

$= 7.2 \times 10^6 m$



Centripetal force = Gravitational force

$m_s w^2 R = \frac{G m_s m_e}{R^2}$

$w^2 R = \frac{G m_e}{R^2}$

$G m_e = w^2 R^3$

but $g R_e^2 = G m_e$

$g R_e^2 = G m_e$

$m_e = \frac{g R_e^2}{G}$

$= \frac{9.81 \times (6.4 \times 10^6)^2}{6.67 \times 10^{-11}}$

$= 6.02 \times 10^{24} kg$

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– Energy required to place satellite in orbit
$(E = P.E + K.E)$

$= GMm\left(\frac{1}{R} - \frac{1}{2R}\right)$

$= GM_e m_s\left(\frac{1}{R} - \frac{1}{2R}\right) = -\frac{GM_e m_s}{2R}$

$= 6.67\times10^{-11} \times 6.02\times10^{24} \times 2000 \left(\frac{1}{6.4\times10^{6}} - \frac{1}{2\times7.2\times10^{6}}\right)$

$= 5.5769\times10^{10}\ J < 6.5\times10^{10}\ J$

Since $6.98\times10^{10} > 6.5\times10^{10}\ J$

–

→ Orbital speed at 600km

$\frac{mv^2}{R} = \frac{GM_s M_e}{R^2}$

$V = \sqrt{\frac{GM}{r}}$

$V = \sqrt{\frac{GM_e}{R}}$

$= \sqrt{\frac{6.67\times10^{-11} \times 6.02\times10^{24}}{7.2\times10^{6}}}$

$\simeq 7.47\times10^{3}\ ms^{-1}$

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Launch Procedure;
- launch vertically initially ✓
- Gain sufficient altitude ✓
- Gradually tilt the rocket horizontally } (02) any 2
- Increase horizontal velocity
- make orbit circular

→ If rocket has already achieved its orbital speed, it will continue orbiting Earth since gravity provides centripetal force.
- If propulsion fails before sufficient horizontal velocity achieved, it will fall back towards Earth.

- $h = \frac{1}{2} g t^2$

$$t = \sqrt{\frac{2h}{g}}$$

$$= \sqrt{\frac{2 \times (8.0 \times 10^5)}{9.81}}$$

=

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Item 2:

- Fission reaction leads to release of energy because from Einstein's mass-energy relation $\Delta E = \Delta m c^2$. Implying change in mass leads to change in energy. So mass of products is slightly less than mass of reactants. This difference in mass (mass defect) accounts for energy released.

— mass of uranium, $U = 235.04394$

— mass of neutron, $n = 1.00874$

— mass of Barium, $Ba = 140.91444$

— Mass of krypton, $Kr = 91.92624$.

→ total mass of reactants;

$235.0439 + 1.0087 = 236.0526$.

⇒ total mass of products;

$140.9144 + 91.9262 + 3(1.0087)$

$= 235.8667$.

→ Mass defect, $\Delta m = (236.0526 - 235.8667)$

$= 0.1859$

Energy released;

Given

$1u = 931 MeV$

$E = (0.1859 \times 931)$

$\simeq 173.0729 MeV$.

Energy in joules

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1 MeV = $1 \times 10^6 \times 1.6 \times 10^{-19}$ J

173.0729 MeV = $(173.0729 \times 1 \times 10^6 \times 1.6 \times 10^{-19})$ J

= $2.7691664 \times 10^{-11}$ J ✓

— Energy from 1kg of uranium;

— mass of one uranium atom = $235.0439 \times 1.66 \times 10^{-27}$

≃ 3.90×10^{-25} kg. ✓

— Number of atoms in 1kg;

$N = \frac{1}{3.90 \times 10^{-25}}$

= 2.56×10^{24} ✓

— total energy = Number of atoms in 1kg × Energy released during fission

= $(2.56 \times 10^{24} \times 2.7691664 \times 10^{-11})$ ✓

= 7.1×10^{13} J

Since $7.1 \times 10^{13} < 7.5 \times 10^{13}$ J, 1kg of pure uranium is not enough to generate electricity needed.

How such systems (nuclear reactor) is controlled;

- Control rods: absorb excess neutrons.
- Moderator: slows neutrons to increase the fission
- Coolant: removes heat.
- Shielding: absorbs harmful radiations
- Containment structure: prevents radioactive materials from escaping

any (3)

- Automatic shut down system controls the chain reaction

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- A neutron is used since it has no charge. Therefore it does not experience electrostatic repulsion from the positively charged uranium nucleus, so it can easily enter and cause fission.
- A Proton or Alpha particle is positively charged and would experience strong electrostatic repulsion.

b). Tube Voltage

$V = 40\,KV = 40{,}000\,V$

K.e, $E = eV$

$= (1.6 \times 10^{-19} \times 40{,}000)$

$= 6.4 \times 10^{-15}\,J$

$E = \frac{hc}{\lambda}$

$\lambda = \frac{hc}{E}$

$= \left(\frac{6.6 \times 10^{-34} \times 3.0 \times 10^{8}}{6.4 \times 10^{-15}}\right)$

$\lambda = 3.09 \times 10^{-11}\,m$

$d = 2.3 \times 10^{-10}\,m$

using Bragg's law:

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$2d\sin\theta = n\lambda$

$n = 2$.

$2 \times 2.3 \times 10^{-10} \sin\theta = 2 \times 3.09 \times 10^{-11}$

$\sin\theta = 0.1345$

$\theta = 7.73^\circ$

required minimum angle is 7°

Since $7.73^\circ > 7^\circ$, the Aluminium sample satisfies the diffraction requirement.

→ How X-rays maintain uniform thickness;

- X-rays are directed through the Aluminium sheet.
- Some are absorbed by aluminium, others transmitted.
- Amount transmitted depends on the thickness.
- A detector measures the transmitted radiation.
- Detector signal is converted into thickness.
- If the sheet becomes too thick, less radiation reaches the detector; if thin more radiation reaches.
- An automatic control system can adjust the production process to maintain constant thickness.

→ Why beta radiation suitable;

- Beta have moderate penetrating power. Therefore they ~~have~~ are suitable for monitoring thickness such as thin metal sheets.

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Alpha - has very low penetration - so unsuitable
Gamma - has very high penetration - so may pass through
thin material with
less useful sensitivity

- Detector placed on opposite side measures beta radiation transmitted through the sheet.

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Item 3:

$A = 60°$

$i = 45°$

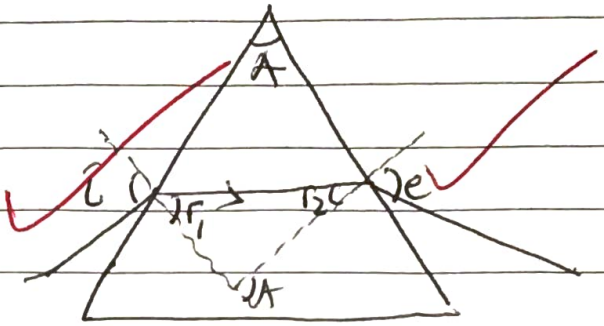
For infrared passing symmetrically:

$i = e = 45°$

also $r_1 + r_2 = A$

$r_1 = r_2$

$r = 30°$



Using Snell's law

$\frac{n_g}{n_a} = \frac{25}{25}$

$n_a \sin i = n_g \sin r_1$

$1 \times \sin 45 = n_{g/a} \cdot \sin 30$

$n_{g/a} = \frac{\sin 45}{\sin 30}$

$= 1.414$

For Ultra Violet radiation

$n_{uv} = 1.525$

$n_a \sin i = n_{g/uv} \sin r$

$\sin 45 = 1.525 \sin r$

$r = \sin^{-1}\left(\frac{\sin 45}{1.525}\right)$

$= 27.62°$

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at second face:

$r_2 = 60 - 27.62$

$r_2 = 32.38°$ ✓

Apply Snell's law:

$n_g \sin r_2 = n_a \sin e$

$1.525 \sin 32.38 = \sin e$ ✓

$e = \sin^{-1}(1.525 \sin 32.38)$

$= 54.74°$ ✓

Deviation of UV

$d = i + e - A$ ✓

$= 45 + 54.74 - 60$

$= 39.74°$ ✓

Deviation for Infrared:

$d = i + e - A$ ✓

$= 45 + 45 - 60$

$= 30°$ ✓

Angular separation $= 39.74 - 30 = 9.74°$ ✓

required separation $= 0.46°$

$9.74° > 0.46°$, the prism can effectively separate Ultra violet and Infrared radiations.

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- 6 plates
- Area of each Plate = $2.0 \times 1.5 = 3m^2$
- total area of 6 plates = $6 \times 3 = 18m^2$

Solar power/constant = $1350 Wm^{-2}$.

Power absorbed;

$P = IA$

$= 1350 \times 18$

$= 24300W$ (Power recieved by the plates).

→ For a black body:

$P = \sigma A T^4$

$$T = \left(\frac{P}{\sigma A}\right)^{\frac{1}{4}}$$

$$= \left(\frac{24300}{5.67 \times 10^{-8} \times 18}\right)^{\frac{1}{4}}$$

$T = 393K$ (temperature generated at the focal spot of the dryer)..

b) The ~~opti~~ optical Pyrometer is better than a bolometer ship since it can measure relatively high temperatures ~~to~~ without physical contact.

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Operation:

- The objective lens is focussed onto the hot object and eye piece adjusted such that the filament is in the same plane with it and the objective.
- Radiation from the hot object is focussed by the eye piece and the Rheostat adjusted such that its intensity merges light produced by the filament. At this point both filament and hot object are at same temperature.
- This temperature is read off from the ammeter initially calibrated to read temperatures directly hence measuring temperature of hot object.

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Item 4:

a) Initially, $m = 32g = 0.032kg$.
- Density of oxygen $= 1.431 kgm^{-3}$.
- $T_1 = 273K$.

- Final temperature, $T_2 = 290K$
- Final volume $V_2 = 0.0166m^3$.
- work done on the gas, $\Delta W = 350J$

Initial volume;

40X½ = 20/20

$$V = \frac{m}{\rho}$$

$$V_1 = \frac{0.032}{1.43}$$

$$V_1 = 0.0224m^3$$

- Since $V_1 > V_2$, the gas has been compressed.

- Temperature increases because its process is extremely rapid, very little heat escapes. Also the work done on the gas increases its internal energy, hence increasing its temperature.

$$\Delta Q = \Delta U + \Delta W$$

$$\Delta Q = 0$$

$$\Delta U = \Delta Q - \Delta W$$

$$\Delta U = 0 - \Delta W$$

- In the scenario, work is done on the gas, not by the gas
- Therefore work done by gas is negative

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- When work is done on the gas → ΔU ↑

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- When work is done by gas → ΔU ↓

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$\Delta W = -350J$ (since compression)

$\Delta U = -(-350) = +350J$ using Convention

$\Delta U = 350J$

Therefore work done on gas leads to increase in ΔU.

— ~~Final Pressure~~; Final Pressure;

Using; $P \cdot V = nRT$

$$P = \frac{nRT}{V}$$

$$P = \frac{1 \times 8.31 \times 290}{0.0166}$$

$$P = 1.45 \times 10^5 Pa$$

Safety limit: $P = 1.64 \times 10^5 Pa$.

Since $1.45 \times 10^5 Pa < 1.64 \times 10^5 Pa$, The pressure is below the safety limit, so the engine is safe from explosion.

b) Projector and wave.

$f = 12cm$

object - screen distance

$D = 10m = 1000cm$

for a projector:

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

$u + v = 1000cm$

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$\frac{1}{f} = \frac{uv}{u+v}$

$\frac{1}{f} = \frac{u+v}{uv}$ $\quad u+v = 1000$

$\frac{1}{12} = \frac{1000}{uv}$

$uv = 12000$ ——— (1)

$u \simeq 987.85\text{cm}$

$v \simeq 12.15\text{ cm}$

∴ The lens should be about 12.15cm from the screen and the slide about 987.9cm from the lens.

$M = \frac{v}{u}$

$= \frac{12.15}{987.85}$

$|M| \simeq 0.0123$

Optical defects

(a) Spherical abberation:

- This is the distortion of image by either a lens or a mirror

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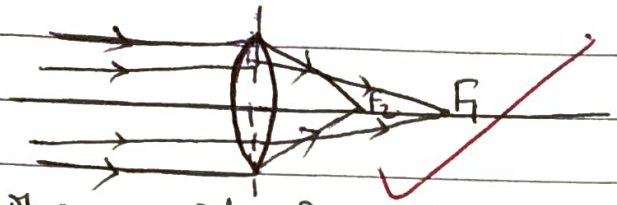
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of wide aperture.



— When a wide of white light is incident on either a lens or a mirror of wide aperture, central rays converge at point far away from the the lens.

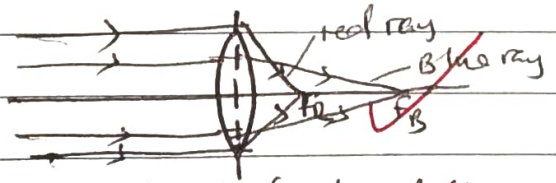
— Rays far from Principal axis converge near the lens. Images formed are ~~formed~~ a series and blurred.

→ In lenses, spherical abberation is minimised by using a stopper ie opaque disc with central hole to cut off marginal rays, allows only paraxial rays.

— In mirrors, we use parabolic mirrors; since they converge wide beam of light incident on its surface to a single focus.

(ii) Chromatic abberation;

— Is the colouring of images produced by a lens.



— When white light is incident on a lens, different colour components are refracted by different amounts. Images corresponding to different colours are formed in different positions along the Principle axis.

→ It can be reduced by using an achromatic doublet. It consists of a convex lens combined with a concave lens

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made from different glass materials. Convex deviates rays and concave nullifies the diversion.

Wave equation;

$y = 0.1\sin(200\pi t - x)$.

Compare with;

$y = A\sin(\omega t - Kx)$

$A = 0.1mm$ ✓

$\omega = 200\pi$

$K = 1rad/m$ ✓

- frequency;

$\omega = 2\pi f$

$200\pi = 2\pi f$

$f = 100Hz$ ✓

- wavelength;

$K = \frac{2\pi}{\lambda}$

$1 = \frac{2\pi}{\lambda}$

$\lambda = 2\pi m$ ✓

- Wave velocity / speed.

$V = f\lambda$ ✓

$= 100 \times 2\pi$

$= 200\pi$ ~~$2.628 m/s$~~.

Modified wave;

twice amplitude. $A_2 = 0.2mm$.

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Double frequency; $f_2 = 200\,Hz$

Therefore $w_2 = 2\pi(200) = 400\pi$

If the wave travels in opposite direction then;

$K_2 = 2\,rad/m$.

$y = 0.2\sin(400\pi t + 2x)$, y is in mm.

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Item 5:

- Motor; 12V, 24W.
- Cabin bulb: 12V, 6W
- Inductor: 2H
- Resistor 470Ω.
- Ac supply: $100V_{r.ms}$, 50Hz.

a) AC supply (Tom's Claim)

- Resistance of the motor;

$P = \frac{V^2}{R}$, $R = \frac{V^2}{P}$

$= \frac{12^2}{24}$ ✓

$= 6\Omega$ ✓

$\frac{1}{2} = \frac{20}{20}$

- Resistance of bulb;

$R = \frac{12^2}{6}$ ✓ $= 24\Omega$ ✓

- Inductive reactance;

$X_L = 2\pi fL$ ✓

$= 2\pi \times 50 \times 2$

$= 628.3\Omega$ ✓

- Total impedance

$R = 24 + 6 + 470 = 500\Omega$.

$Z = \sqrt{R^2 + X_L^2}$ ✓

$= \sqrt{(500)^2 + (628.3)^2}$

$= 803\Omega$ ✓

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→ Circuit current;

$$I = \frac{V}{Z}$$

$$I = \frac{100}{803}$$

$$I \approx 0.125A \quad 0.1245A$$

→ Voltage across coil;

$$V_L = I X_L$$

$$V = 0.125 \times 628.3$$

$$V \simeq 78.2V$$

- So the coil voltage is much greater than 8V.

→ Why the motor fails;

Voltage across the motor

$$V = IR$$

$$= 0.125 \times 6$$

$$\simeq 0.747V < 12V$$

- Motor relieves far too little voltage, it cannot operate normally.

6) D.C Supply (Sarah's claim)

- If 100V DC source is used and switch S_2 remains open
- Inductor behaves like a short circuit
- $R_T = 470 + 6 + 24$

$$= 500\Omega$$

$$I = \frac{100}{500} = 0.20A$$

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- Shunt is low-resistance resistor connected in parallel with a device to provide an alternative path for current.

→ When D.C is switched on, inductor opposes ↑I when D.C flows for a long time current is constant $\frac{dI}{dt} = 0$, so no more voltage in inductor with A.C it is continuous.

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Voltage across bulb:

$V = IR_b$

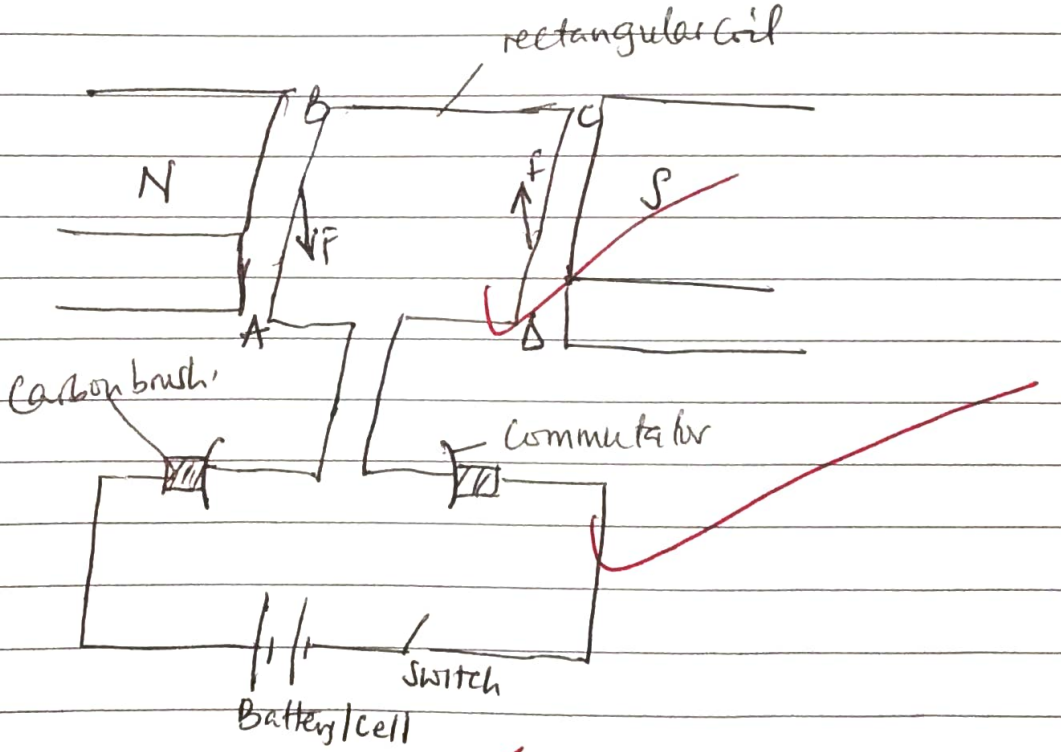
$= 0.2 \times 24$

$= 4.8V < 12V$, the rating of the bulb.

- bulb will not blow, it will be under powered and will produce dim lights. So a shunt is not necessary to protect it under these assumptions.

~~Operation of D.C motor used to run the fan:~~

- the device used to run the fan is a d.c motor.



- Current flows in the direction ABCD, when switch is closed. AB experiences downward force, CD upwards force, the two forces form a couple, as it rotates in anti clockwise.
- As the coil rotates, ~~direction~~ Commutator loose contact

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~~with from~~ with Carbon brushes and interchange positions
- direction of forces changes but direction of rotation remains constant.

ITEM 6:

i) Protecting galvanometer;

$N = 500$, $A = 4.0 \times 10^{-4} m^2$, $B = 0.25 T$, $K = 3.0 \times 10^{-5} Nm rad^{-1}$

$\theta = 0.4 rad$.

Torque on the coil;

$T = NBIA$

at equilibrium;

$NBIA = K\theta$

$$I = \frac{K\theta}{NBA} = \frac{3.0 \times 10^{-5} \times 0.4}{500 \times 0.25 \times 4.0 \times 10^{-4}}$$

$I = 2.4 \times 10^{-4} A$

$\simeq 0.24 mA$ (Current corresponding to 0.4 rad deflection).

- Resistance of the protective wire;

$\rho = 1.7 \times 10^{-8} \Omega m$, $L = 0.5 m$, $A = 4.25 \times 10^{-10} m^2$

$$R = \frac{\rho L}{A}$$

$$R = \frac{1.7 \times 10^{-8} \times 0.5}{4.25 \times 10^{-10}}$$

$R = 20 \Omega$

20/20

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- Power dissipated in the wire;

$P = I^2 R$

$= (2.4 \times 10^{-4})^2 \times 20$

$= 1.152 \times 10^{-6}$ W

$E = Pt$

$= (1.152 \times 10^{-6} \times 30)$

$= 3.456 \times 10^{-5}$ J

The limit is 1.62×10^{-5} J

Since $3.456 \times 10^{-5} > 1.62 \times 10^{-5}$ J; Protective wire would generate more heat than stated amount, therefore technician should not allow this operation.

- time to reach limit;

$t = E/P$

$= \dfrac{1.62 \times 10^{-5}}{1.152 \times 10^{-6}}$

$\simeq 14.1$s

- So stated threshold would be reached after about 14s

b) Radio tuning circuit

- Inductance of solenoid

$N = 600$, $l = 22.7$cm $= 0.227$m, $r = 1.42$cm $= 0.0142$m

$A = \pi r^2$

$= \pi (0.0142)^2$

$\simeq 6.33 \times 10^{-4}$ m^2

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$L = \frac{N\Phi}{I}$ $\Phi = BA$ $B = \mu_0 n I$ n = no of turns per metre $\Phi = \frac{\mu_0 n I A \cdot N}{I}$

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$L = \frac{\mu_0 N^2 A}{L}$

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For Solenoid;

$$L = \frac{\mu_0 N^2 A}{L}$$

$\mu_0 = 4\pi \times 10^{-7} Hm^{-1}$

$= \frac{4\pi \times 10^{-7} \times (600)^2 \times 6.33 \times 10^{-4}}{0.0227}$

$L \simeq 1.26 \times 10^{-3} H$

$\simeq 1.26 mH$

– Required capacator for 2.516 MHz.

Resonance occurs when;

$$f = \frac{1}{2\pi\sqrt{LC}}$$

$$C = \frac{1}{(2\pi f)^2 L}$$

$f = 2.516 \times 10^6 Hz$, $L = 1.26 \times 10^{-3} H$

$$C = \frac{1}{(2\pi \times 2.516 \times 10^6)^2 \times (1.26 \times 10^{-3})}$$

$C \simeq 3.17 \times 10^{-12} F$

$\simeq 3.17 pF$

– Galvanometer doesnot deflect because the circuit is being supplied with $f = 40Hz$, but it has been tuned for 2.516 MHz. These two frequencies are different since at 40Hz, capacitor's reactance is extremely large.

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Impedance becomes resistive when resistance, R is much greater than its reactance.

$Z = \sqrt{R^2 + X_L^2}$

$R \gg X_L \rightarrow Z \simeq R$

Circuit behaves like an ordinary resistor.

- Impedance - total opposition to flow of an AC in circuit having bulb, resistors, capacitors and inductor.

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$X_C = \frac{1}{2\pi f C} = \frac{1}{2\pi \times 40 \times 3.17 \times 10^{-12}} = 1.26 \times 10^9\ \Omega$

$I = \frac{12}{1.26 \times 10^9}$

$= 9.5 \times 10^{-9}\ A$

$\simeq 9.5\ nA$, current is extremely small.

- Expected galvanometer deflection:

$I = \frac{k\theta}{NBA}$

$\theta = \frac{NBA\,I}{k}$

$= \frac{500 \times 0.25 \times 4.0 \times 10^{-4} \times 9.5 \times 10^{-9}}{3.0 \times 10^{-5}}$

$\simeq 1.6 \times 10^{-5}$ rad (which is very small and essentially zero on the galvanometer scale).

- How circuit should be modified?
- For the circuit to resonate with the incoming radio signal.

$f_{source} = 2.516\ MHz$ and the capacitor should be approximately 3.17pF. Such that at resonance $X_L = X_C$ and impedance becomes mainly resistive allowing much large current.

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